Parshvanath Charitable Trust's
A. P. SHAH INSTITUTE OF TECHNOLOGY
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(Religious Jain Minority)

Department of Computer Engineering

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Class / Branch: S.E / COMP
Subject: Analysis of Algorithm
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Experiment No.2

Aim:Implement Quick sort algorithm using Divide & Conquer approach Program:

```c
#include <stdio.h>

void quicksort(int a[], int first, int last)
{
   int i, j, pivot, temp;

   if (first < last)
   {
      pivot = first;
      i = first+ 1;
      j = last;

      while (i <= j)
     {

         while (i <= j && a[i] <= a[pivot])
         {
           i++;

         }

         while (i <= j && a[j] > a[pivot])
         {
           j--;
         }

         if (i < j)
         {
           temp = a[i];
           a[i] = a[j];
           a[j] = temp;
         }
      }

      temp = a[pivot];
      a[pivot] = a[j];
      a[j] = temp;

      quicksort(a, first, j - 1);
      quicksort(a, j + 1, last);
   }
}

int main()
{
      quicksort(a, j + 1, last);
   }
}

int main()
{
      int a[100], n, i;

      printf("Enter number of elements: ");
      scanf("%d", &n);

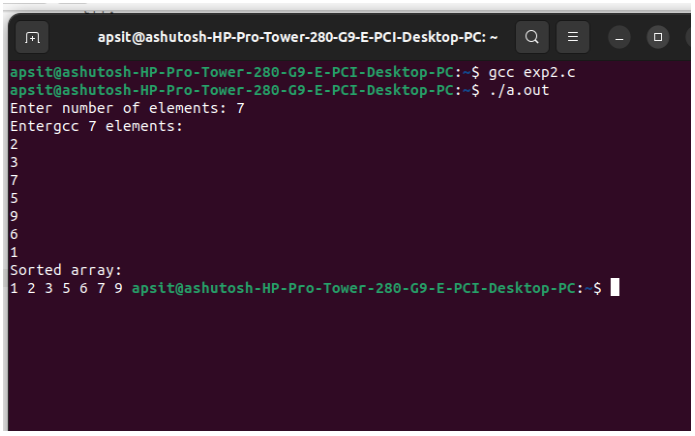
      printf("Entergcc %d elements: \n", n);
      for ( i = 0 ;i < n; i++)
        scanf("%d", &a[i]);

      quicksort(a, 0, n -1);

      printf("Sorted array:\n");
      for ( i = 0; i < n; i++)
        printf("%d ", a[i]);

      return 0;
}
```

Output:



Conclusion:Quick Sort is one of the fastest sorting algorithms in practice, especially when implemented in-place. Using the first element as pivot is simple, but can lead to worst-case time complexity on already sorted or reverse-sorted data. Still, it's widely used due to its efficiency, low memory usage, and ease of implementation.

Exercise 1: Aim:Implement Insertion sort algorithm & analyse its Time & Space Complexity
Program:

```c
#include <stdio.h>

int main() {
    int n, i, j, min, temp;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter elements:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    for(i = 0; i < n - 1; i++) {
        min = i;

        for(j = i + 1; j < n; j++) {
            if(arr[j] < arr[min]) {
                min = j;
            }
        }

        temp = arr[i];
        arr[i] = arr[min];
        arr[min] = temp;
    }

    printf("Sorted array:\n");
    for(i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }

    return 0;
}
```

Output:-

```
PS C:\Coding_c> cd "c:\Coding_c\" ; if ($?) { gcc exp4.c -o exp4 } ; if ($?) { .\exp4 }
Enter number of elements: 5
Enter elements:
10
3
8
35
1
Sorted array:
1 3 8 10 35
PS C:\Coding_c>
```